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WO-200-01 — Internal inspection of vertical vessel C-101

Area	A200 Vessels and column	**Task**	VI Internal inspection
Priority	2 (integrity)	**Issued by**	Facility owner (demonstration)
Equipment	C-101	**Lines**	4"-P300-P-1010, 6"-P300-P-1011, 6"-P300-P-1017, 3"-P300-P-1015, 1"-P150-V-6011, 1"-P150-D-6012, 1"-P150-N-3013, 3/4"-P300-I-6013, 3/4"-P300-I-6014, 3/4"-P300-SP-6015
Isometrics	ISO-4"-P300-P-1010, ISO-6"-P300-P-1011, ISO-6"-P300-P-1017, ISO-3"-P300-P-1015, ISO-1"-P150-V-6011, ISO-1"-P150-D-6012, ISO-1"-P150-N-3013, ISO-3/4"-P300-I-6013, ISO-3/4"-P300-I-6014, ISO-3/4"-P300-SP-6015		
Permits	General work permit, Confined-space entry permit		

Reason

Scheduled internal inspection of the column.

Scope of work

1. Isolate C-101 per the isolation list; drain, vent, purge and gas-test for entry.
2. Open for entry; clean as needed for inspection.
3. Internal visual examination; ultrasonic thickness at the 16 TMLs.
4. Close with new gaskets on every opened joint; remove blinds; restore the line-up.

Affected items: C-101

Rules: SW-ITP-1, SW-ITP-2, SW-ITP-3, SW-TEST-2, SW-WO-1, SW-WO-2, SW-WO-3, SW-WO-5, SW-WO-6

Isolation list (SW-WO-3)

#	Kind	Tag	Line	Action	Location	Fitted ✓	Removed ✓
1	valve	4"-P300-P-1010/B2	4"-P300-P-1010	Close and lock VALVE, GATE, WEDGE, OS&Y 4" P300 (4"-P300-P-1010/B2) on C-101-N1	E 68'-6" N 21'-8-3/16" EL 116'-0"		
2	blind	C-101-N1	4"-P300-P-1010	Fit a blind between 4"-P300-P-1010/B2 and the nozzle flange C-101-N1	E 68'-6" N 22'-3-15/16" EL 116'-0"		
3	valve	6"-P300-P-1011/A2	6"-P300-P-1011	Close and lock VALVE, GATE, WEDGE, OS&Y 6" P300 (6"-P300-P-1011/A2) on C-101-N2	E 68'-6" N 25'-0" EL 127'-11-11/16"		
4	blind	C-101-N2	6"-P300-P-1011	Fit a blind between 6"-P300-P-1011/A2 and the nozzle flange C-101-N2	E 68'-6" N 25'-0" EL 127'-4-9/16"		
5	valve	6"-P300-P-1017/A2	6"-P300-P-1017	Close and lock VALVE, GATE, WEDGE, OS&Y 6" P300 (6"-P300-P-1017/A2) on C-101-N3	E 68'-6" N 21'-8-9/16" EL 106'-6"		
6	blind	C-101-N3	6"-P300-P-1017	Fit a blind between 6"-P300-P-1017/A2 and the nozzle flange C-101-N3	E 68'-6" N 22'-3-3/4" EL 106'-6"		
7	valve	3"-P300-P-1015/B2	3"-P300-P-1015	Close and lock VALVE, GATE, WEDGE, OS&Y 3" P300 (3"-P300-P-1015/B2) on C-101-N4	E 68'-6" N 21'-9-3/4" EL 125'-0"		
8	blind	C-101-N4	3"-P300-P-1015	Fit a blind between 3"-P300-P-1015/B2 and the nozzle flange C-101-N4	E 68'-6" N 22'-4-11/16" EL 125'-0"		
9	valve	1"-P150-V-6011/2	1"-P150-V-6011	Close and lock VALVE, BALL, FLOATING 1" P150 (1"-P150-V-6011/2) on C-101-N5	E 70'-0" N 25'-0" EL 127'-4-13/16"		
10	blind	C-101-N5	1"-P150-V-6011	Fit a blind between 1"-P150-V-6011/2 and the nozzle flange C-101-N5	E 70'-0" N 25'-0" EL 126'-9-15/16"		
11	valve	1"-P150-D-6012/2	1"-P150-D-6012	Close and lock VALVE, BALL, FLOATING 1" P150 (1"-P150-D-6012/2) on C-101-N6	E 65'-5-1/2" N 25'-0" EL 106'-0"		
12	blind	C-101-N6	1"-P150-D-6012	Fit a blind between 1"-P150-D-6012/2 and the nozzle flange C-101-N6	E 66'-3/8" N 25'-0" EL 106'-0"		
13	valve	1"-P150-N-3013/B2	1"-P150-N-3013	Close and lock VALVE, BALL, FLOATING 1" P150 (1"-P150-N-3013/B2) on C-101-N7	E 68'-6" N 28'-1/2" EL 115'-0"		
14	blind	C-101-N7	1"-P150-N-3013	Fit a blind between 1"-P150-N-3013/B2 and the nozzle flange C-101-N7	E 68'-6" N 27'-5-11/16" EL 115'-0"		
15	valve	3/4"-P300-I-6013/2	3/4"-P300-I-6013	Close and lock VALVE, BALL, FLOATING 3/4" P300 (3/4"-P300-I-6013/2) on C-101-N8	E 65'-5-3/4" N 25'-0" EL 108'-6"		
16	blind	C-101-N8	3/4"-P300-I-6013	Fit a blind between 3/4"-P300-I-6013/2 and the nozzle flange C-101-N8	E 66'-1/2" N 25'-0" EL 108'-6"		
17	valve	3/4"-P300-I-6014/2	3/4"-P300-I-6014	Close and lock VALVE, BALL, FLOATING 3/4" P300 (3/4"-P300-I-6014/2) on C-101-N9	E 68'-6" N 28'-1/4" EL 117'-6"		
18	blind	C-101-N9	3/4"-P300-I-6014	Fit a blind between 3/4"-P300-I-6014/2 and the nozzle flange C-101-N9	E 68'-6" N 27'-5-9/16" EL 117'-6"		
19	valve	3/4"-P300-SP-6015/2	3/4"-P300-SP-6015	Close and lock VALVE, BALL, FLOATING 3/4" P300 (3/4"-P300-SP-6015/2) on C-101-N10	E 65'-5-3/4" N 25'-0" EL 107'-6"		
20	blind	C-101-N10	3/4"-P300-SP-6015	Fit a blind between 3/4"-P300-SP-6015/2 and the nozzle flange C-101-N10	E 66'-1/2" N 25'-0" EL 107'-6"		

Inspections

Id	Inspection	Criteria	Basis
WO-200-01-I1	Internal visual examination of the shell, heads and nozzles	Findings recorded against the owner's integrity criteria	SW-NDE-4

Id	Inspection	Criteria	Basis
WO-200-01-I2	Ultrasonic thickness at the 16 TMLs listed	Each reading above the owner's minimum wall	SW-WO-6

Thickness monitoring locations (SW-WO-6)

TML	Item	Location	Position	Nominal wall	R1	R2	R3	R4
WO-200-01-TML01	C-101	Nozzle neck C-101-N1	E 68'-6" N 22'-3-15/16" EL 116'-0"		top: ____	side: ____	bottom: ____	side: ____
WO-200-01-TML02	C-101	Nozzle neck C-101-N2	E 68'-6" N 25'-0" EL 127'-4-9/16"		top: ____	side: ____	bottom: ____	side: ____
WO-200-01-TML03	C-101	Nozzle neck C-101-N3	E 68'-6" N 22'-3-3/4" EL 106'-6"		top: ____	side: ____	bottom: ____	side: ____
WO-200-01-TML04	C-101	Nozzle neck C-101-N4	E 68'-6" N 22'-4-11/16" EL 125'-0"		top: ____	side: ____	bottom: ____	side: ____
WO-200-01-TML05	C-101	Nozzle neck C-101-N5	E 70'-0" N 25'-0" EL 126'-9-15/16"		top: ____	side: ____	bottom: ____	side: ____
WO-200-01-TML06	C-101	Nozzle neck C-101-N6	E 66'-3/8" N 25'-0" EL 106'-0"		top: ____	side: ____	bottom: ____	side: ____
WO-200-01-TML07	C-101	Nozzle neck C-101-N7	E 68'-6" N 27'-5-11/16" EL 115'-0"		top: ____	side: ____	bottom: ____	side: ____
WO-200-01-TML08	C-101	Nozzle neck C-101-N8	E 66'-1/2" N 25'-0" EL 108'-6"		top: ____	side: ____	bottom: ____	side: ____
WO-200-01-TML09	C-101	Nozzle neck C-101-N9	E 68'-6" N 27'-5-9/16" EL 117'-6"		top: ____	side: ____	bottom: ____	side: ____
WO-200-01-TML10	C-101	Nozzle neck C-101-N10	E 66'-1/2" N 25'-0" EL 107'-6"		top: ____	side: ____	bottom: ____	side: ____
WO-200-01-TML11	C-101	Bottom head centre	E 68'-6" N 25'-0" EL 103'-8"		centre: ____			
WO-200-01-TML12	C-101	Top head centre	E 68'-6" N 25'-0" EL 125'-8-1/2"		centre: ____			
WO-200-01-TML13	C-101	Shell at 0.000" above the lower tangent	E 68'-6" N 25'-0" EL 104'-8-1/4"		0°: ____	90°: ____	180°: ____	270°: ____
WO-200-01-TML14	C-101	Shell at 6'-0" above the lower tangent	E 68'-6" N 25'-0" EL 110'-8-1/4"		0°: ____	90°: ____	180°: ____	270°: ____
WO-200-01-TML15	C-101	Shell at 12'-0" above the lower tangent	E 68'-6" N 25'-0" EL 116'-8-1/4"		0°: ____	90°: ____	180°: ____	270°: ____
WO-200-01-TML16	C-101	Shell at 18'-0" above the lower tangent	E 68'-6" N 25'-0" EL 122'-8-1/4"		0°: ____	90°: ____	180°: ____	270°: ____

Test — Service leak test

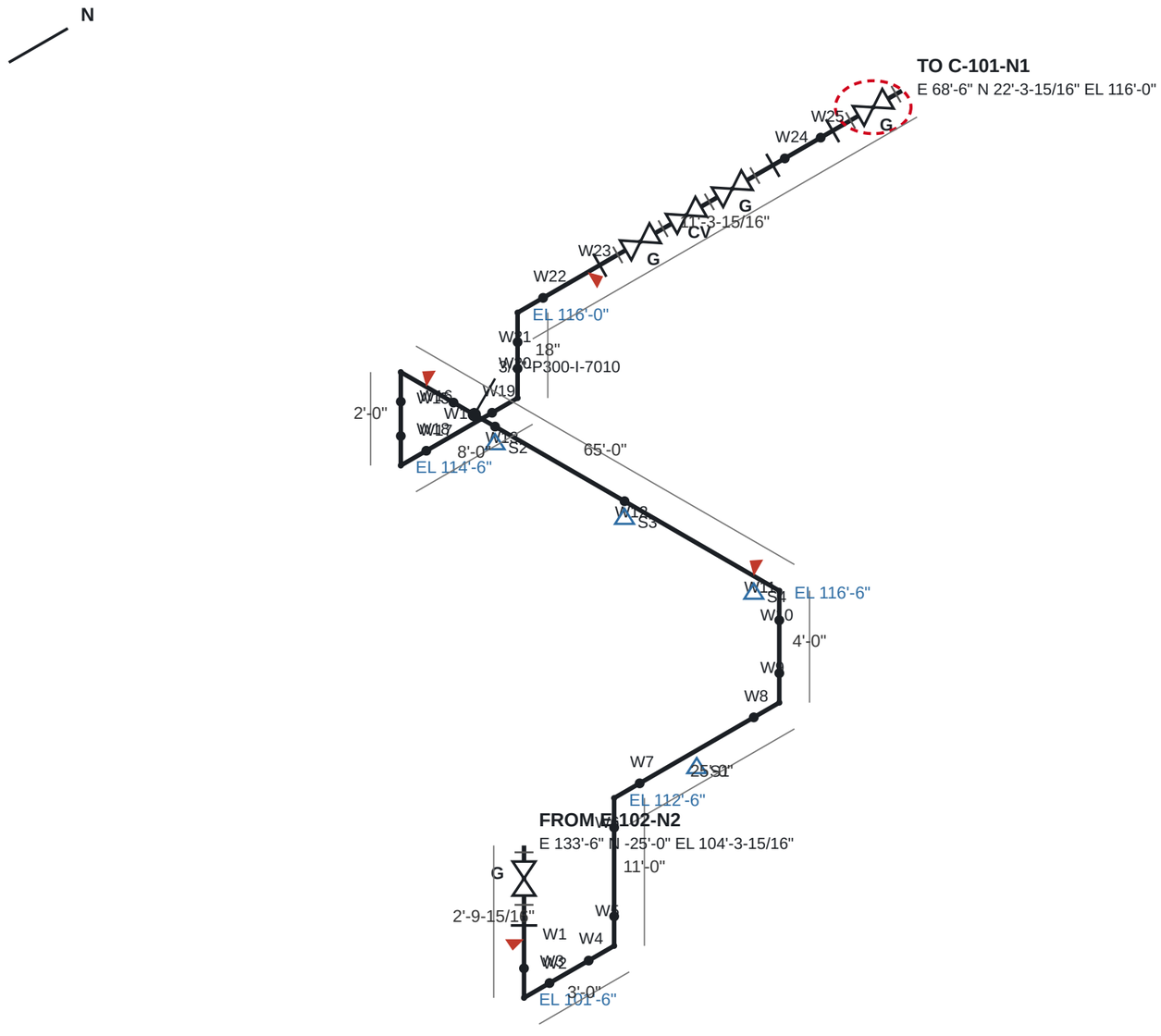
Description	Every opened joint checked at operating pressure on start-up
Basis	SW-TEST-2

Inspection and test plan (SW-ITP-1..3)

H hold · W witness · R review · S surveillance. Status: accepted / rejected / not applicable.

#	Activity	Reference	Acceptance	Record	Contr.	Owner	3rd party	Status	Signed (contr. / owner / 3rd)	Date	Remark
1	Work permit issued; isolation list (valves locked, blinds fitted) verified at zero pressure and drained	SW-WO-3, isolation list	Every isolation in place and tagged; zero pressure at the vent	Signed isolation certificate	H	H	—				
2	Equipment vented, purged and gas-tested for entry; confined-space entry permit issued	Owner entry procedure	Atmosphere safe for entry per owner limits	Gas test record, entry permit	H	H	—				
3	Opened for entry; cleaned for inspection	Owner procedure	Surfaces clean enough to inspect	Daily report	S	S	—				
4	Internal visual examination	SW-NDE-4	Findings recorded	Inspection report with photographs	W	H	W				
5	Ultrasonic thickness at 16 TMLs	SW-WO-6	Above the owner's minimum wall	Thickness record per TML	S	R	R				
6	Owner's decision on findings (repair, monitor or accept)	Owner integrity procedure	Decision recorded	Signed disposition	R	H	R				
7	Closed with new gaskets on every opened joint; studs to preload	SW-WO-5	Preload as stated	Bolt-up record	W	W	—				
8	Service leak test of opened joints on start-up	SW-TEST-2	No leak	Leak test record	W	W	—				
9	Blinds removed, isolations returned to normal line-up per the isolation list	SW-WO-3	Every blind removed and accounted for	Isolation certificate closed	H	H	—				
10	Records complete; work order handed back to operations	SW-ITP-3	All ITP activities accepted or not applicable	Signed ITP and completion record	H	H	R				

ISOMETRIC — 4"-P300-P-1010 — Internal inspection of vertical vessel C-101



BILL OF MATERIALS

1	GASKET 4" P300 COMPRESSED SHEET	8
2	VALVE, GATE, WEDGE, OS&Y 4" P300	4
3	STUD SET 4 × 1" × 6-1/4" WITH 2 NUTS, FOR 4" P300	8
4	FLANGE WELD-NECK 4" P300 RAISED FACE	4
5	PIPE 4" P300 BEVELLED ENDS	114'-9-1/8"
6	ELBOW 90° R = 5/3 OD 4" P300	9
7	TEE REDUCING 4" × 3/4" P300	1
8	VALVE, CONTROL, GLOBE WITH DIAPHRAGM ACTUATOR 4" P300	1

WELDS 25 (4 FIELD) · SUPPORTS 4 · SPOOLS 6

W1F W2 W3 W4 W5 W6 W7 W8 W9 W10
W11F W12 W13 W14 W15F W16 W17 W18 W19 W20
W21 W22 W23F W24 W25

4"-P300-P-1010

CLASS H300 — Hydrocarbon process
SERVICE P · 4" · P300 · DESIGN 300 psi
E-102-N2 → C-101-N1
PIPE 114'-9-1/8" · WEIGHT 1886 lb

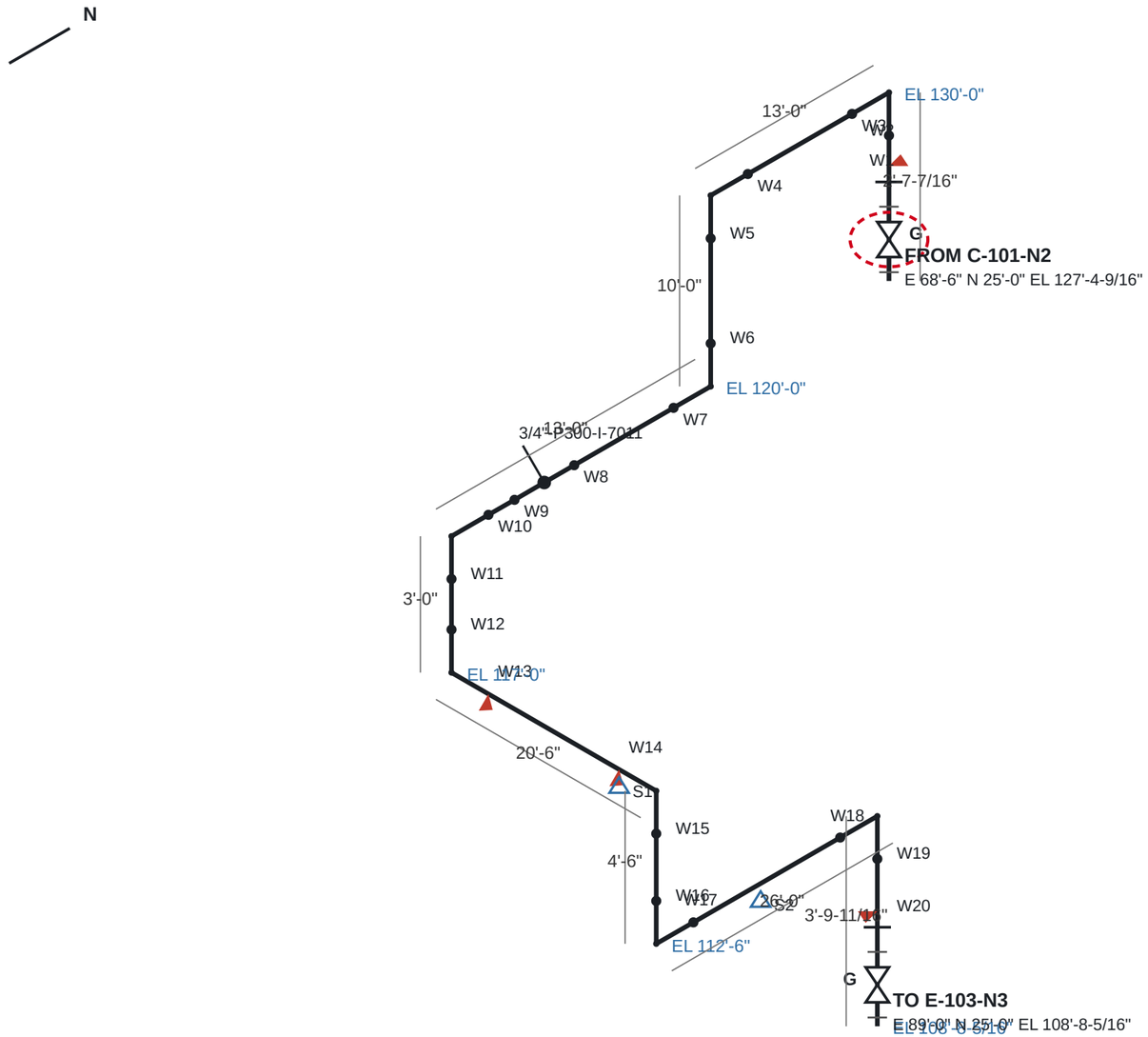
DWG ISO-4"-P300-P-1010 REV 0 WO-200-01

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Isometric ISO-6"-P300-P-1011

ISOMETRIC — 6"-P300-P-1011 — Internal inspection of vertical vessel C-101



BILL OF MATERIALS

1	GASKET 6" P300 COMPRESSED SHEET	4
2	VALVE, GATE, WEDGE, OS&Y 6" P300	2
3	STUD SET 8 × 0.625" × 4" WITH 2 NUTS, FOR 6" P300	4
4	FLANGE WELD-NECK 6" P300 RAISED FACE	2
5	PIPE 6" P300 BEVELLED ENDS	79'-2-13/16"
6	ELBOW 90° R = 5/3 OD 6" P300	8
7	TEE REDUCING 6" × 3/4" P300	1

WELDS 20 (4 FIELD) · SUPPORTS 2 · SPOOLS 5

W1F W2 W3 W4 W5 W6 W7 W8 W9 W10
W11 W12 W13F W14F W15 W16 W17 W18 W19 W20F

6"-P300-P-1011

CLASS H300 — Hydrocarbon process
SERVICE P · 6" · P300 · DESIGN 300 psi
C-101-N2 → E-103-N3
PIPE 79'-2-13/16" · WEIGHT 1677 lb

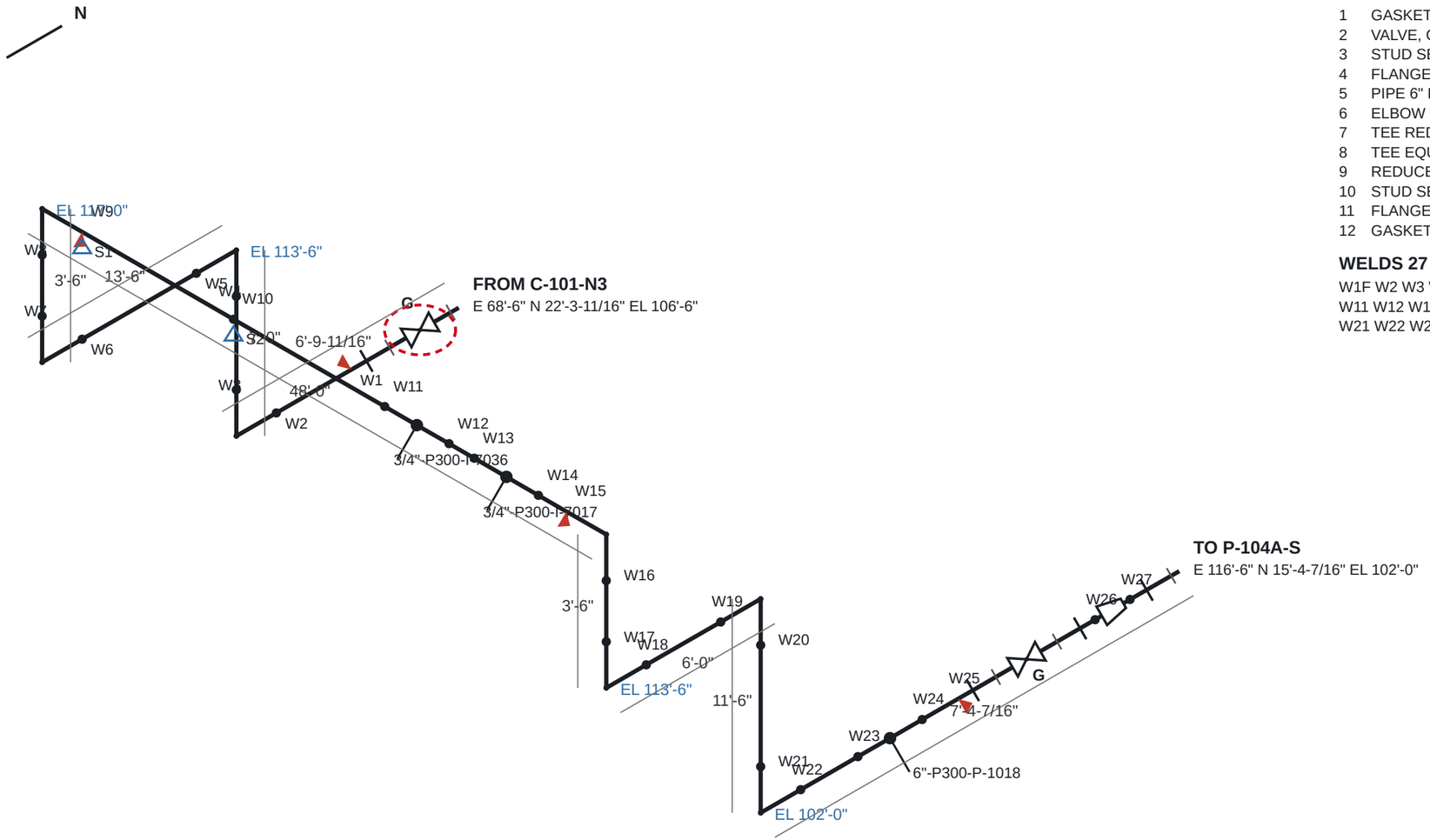
DWG ISO-6"-P300-P-1011 REV 0 WO-200-01

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Isometric ISO-6"-P300-P-1017

ISOMETRIC — 6"-P300-P-1017 — Internal inspection of vertical vessel C-101



BILL OF MATERIALS

1	GASKET 6" P300 COMPRESSED SHEET	4
2	VALVE, GATE, WEDGE, OS&Y 6" P300	2
3	STUD SET 8 × 0.625" × 4" WITH 2 NUTS, FOR 6" P300	4
4	FLANGE WELD-NECK 6" P300 RAISED FACE	3
5	PIPE 6" P300 BEVELLED ENDS	86'-11-5/8"
6	ELBOW 90° R = 5/3 OD 6" P300	8
7	TEE REDUCING 6" × 3/4" P300	2
8	TEE EQUAL 6" P300	1
9	REDUCER CONCENTRIC 6" × 4" P300	1
10	STUD SET 4 × 1" × 6-1/4" WITH 2 NUTS, FOR 4" P300	1
11	FLANGE WELD-NECK 4" P300 RAISED FACE	1
12	GASKET 4" P300 COMPRESSED SHEET	1

WELDS 27 (4 FIELD) · SUPPORTS 2 · SPOOLS 6

W1F W2 W3 W4 W5 W6 W7 W8 W9F W10
W11 W12 W13 W14 W15F W16 W17 W18 W19 W20
W21 W22 W23 W24 W25F W26 W27

6"-P300-P-1017

CLASS H300 — Hydrocarbon process
SERVICE P · 6" · P300 · DESIGN 300 psi
C-101-N3 → P-104A-S
PIPE 86'-11-5/8" · WEIGHT 1867 lb

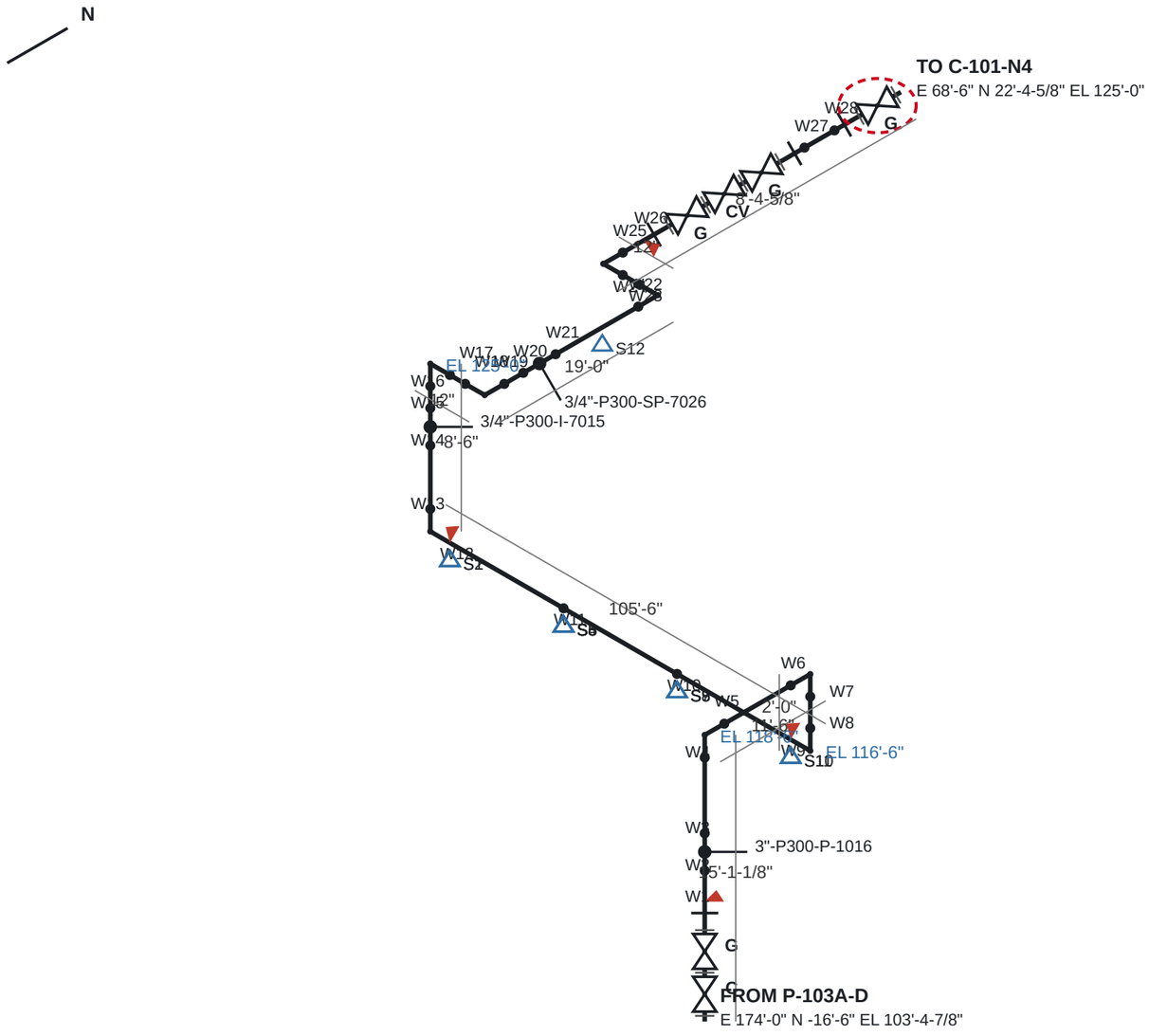
DWG ISO-6"-P300-P-1017 REV 0 WO-200-01

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Isometric ISO-3"-P300-P-1015

ISOMETRIC — 3"-P300-P-1015 — Internal inspection of vertical vessel C-101



BILL OF MATERIALS

1	GASKET 3" P300 COMPRESSED SHEET	9
2	VALVE, SWING CHECK 3" P300	1
3	STUD SET 4 × 0.875" × 5-3/4" WITH 2 NUTS, FOR 3" P300	9
4	VALVE, GATE, WEDGE, OS&Y 3" P300	4
5	FLANGE WELD-NECK 3" P300 RAISED FACE	4
6	PIPE 3" P300 BEVELLED ENDS	155'-4-3/8"
7	TEE EQUAL 3" P300	1
8	ELBOW 90° R = 5/3 OD 3" P300	8
9	TEE REDUCING 3" × 3/4" P300	2
10	VALVE, CONTROL, GLOBE WITH DIAPHRAGM ACTUATOR 3" P300	1

WELDS 28 (4 FIELD) · SUPPORTS 12 · SPOOLS 6

W1F W2 W3 W4 W5 W6 W7 W8 W9F W10
W11 W12F W13 W14 W15 W16 W17 W18 W19 W20
W21 W22 W23 W24 W25 W26F W27 W28

3"-P300-P-1015

CLASS H300 — Hydrocarbon process
SERVICE P · 3" · P300 · DESIGN 300 psi
P-103A-D → C-101-N4
PIPE 155'-4-3/8" · WEIGHT 1731 lb

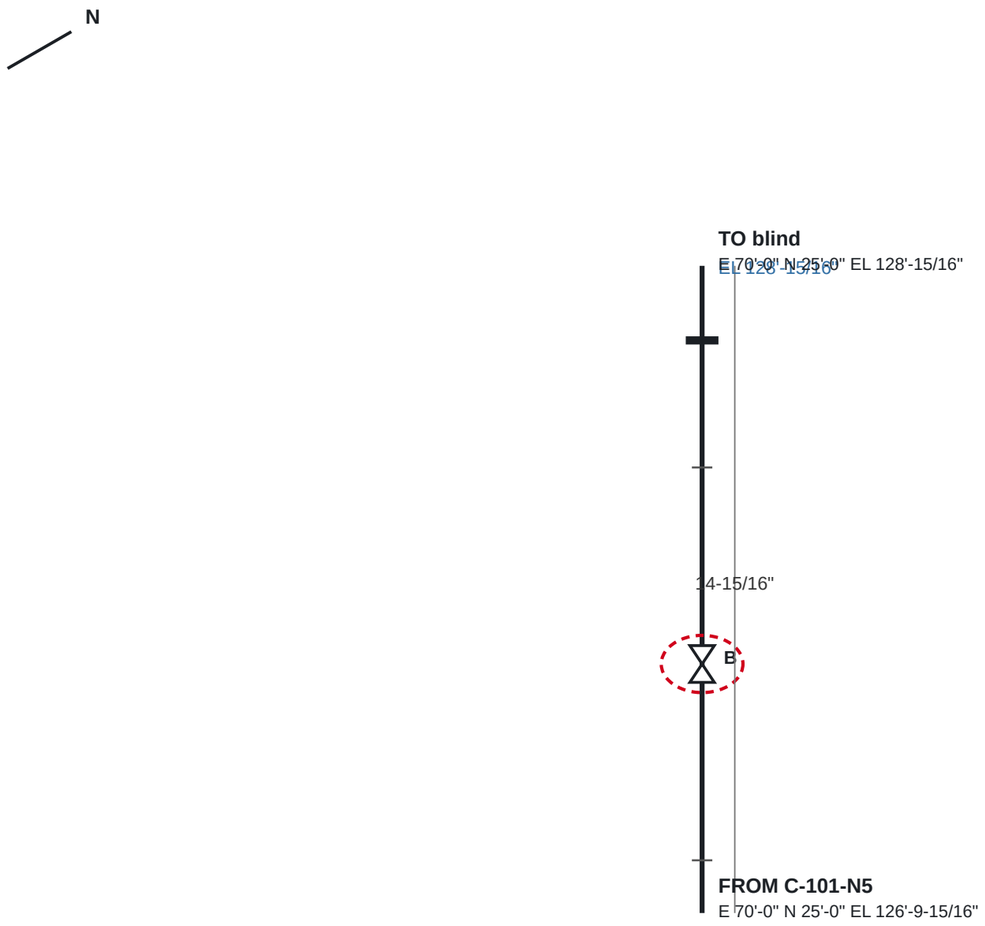
DWG ISO-3"-P300-P-1015 REV 0 WO-200-01

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Isometric ISO-1"-P150-V-6011

ISOMETRIC — 1"-P150-V-6011 — Internal inspection of vertical vessel C-101



BILL OF MATERIALS

1	GASKET 1" P150 COMPRESSED SHEET	2
2	VALVE, BALL, FLOATING 1" P150	1
3	STUD SET 4 × 0.5" × 4" WITH 2 NUTS, FOR 1" P150	2
4	FLANGE BLIND 1" P150 RAISED FACE	1

WELDS 0 (0 FIELD) · SUPPORTS 0 · SPOOLS 0

1"-P150-V-6011

CLASS H150 — Hydrocarbon process
SERVICE V · 1" · P150 · DESIGN 150 psi
C-101-N5 → blind
PIPE 0.000" · WEIGHT 39 lb

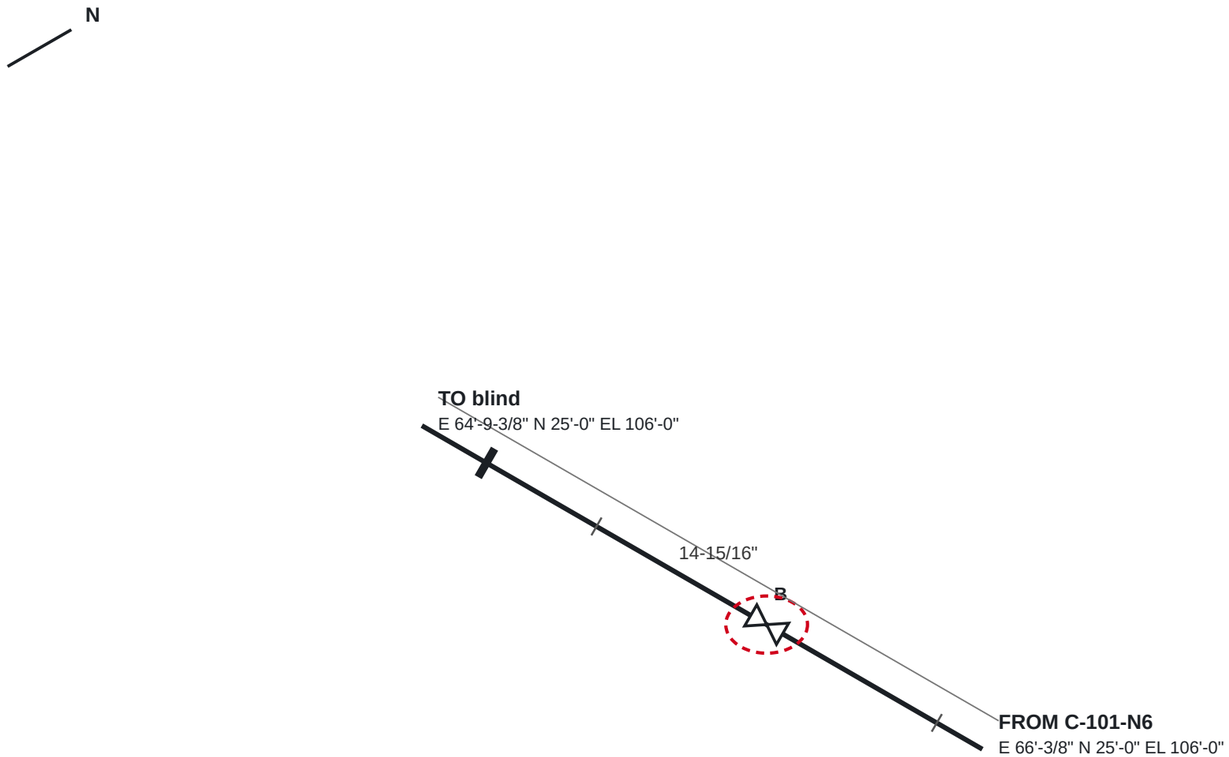
DWG ISO-1"-P150-V-6011 REV 0 WO-200-01

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Isometric ISO-1"-P150-D-6012

ISOMETRIC — 1"-P150-D-6012 — Internal inspection of vertical vessel C-101



BILL OF MATERIALS

1	GASKET 1" P150 COMPRESSED SHEET	2
2	VALVE, BALL, FLOATING 1" P150	1
3	STUD SET 4 × 0.5" × 4" WITH 2 NUTS, FOR 1" P150	2
4	FLANGE BLIND 1" P150 RAISED FACE	1

WELDS 0 (0 FIELD) · SUPPORTS 0 · SPOOLS 0

1"-P150-D-6012

CLASS H150 — Hydrocarbon process
SERVICE D · 1" · P150 · DESIGN 150 psi
C-101-N6 → blind
PIPE 0.000" · WEIGHT 39 lb

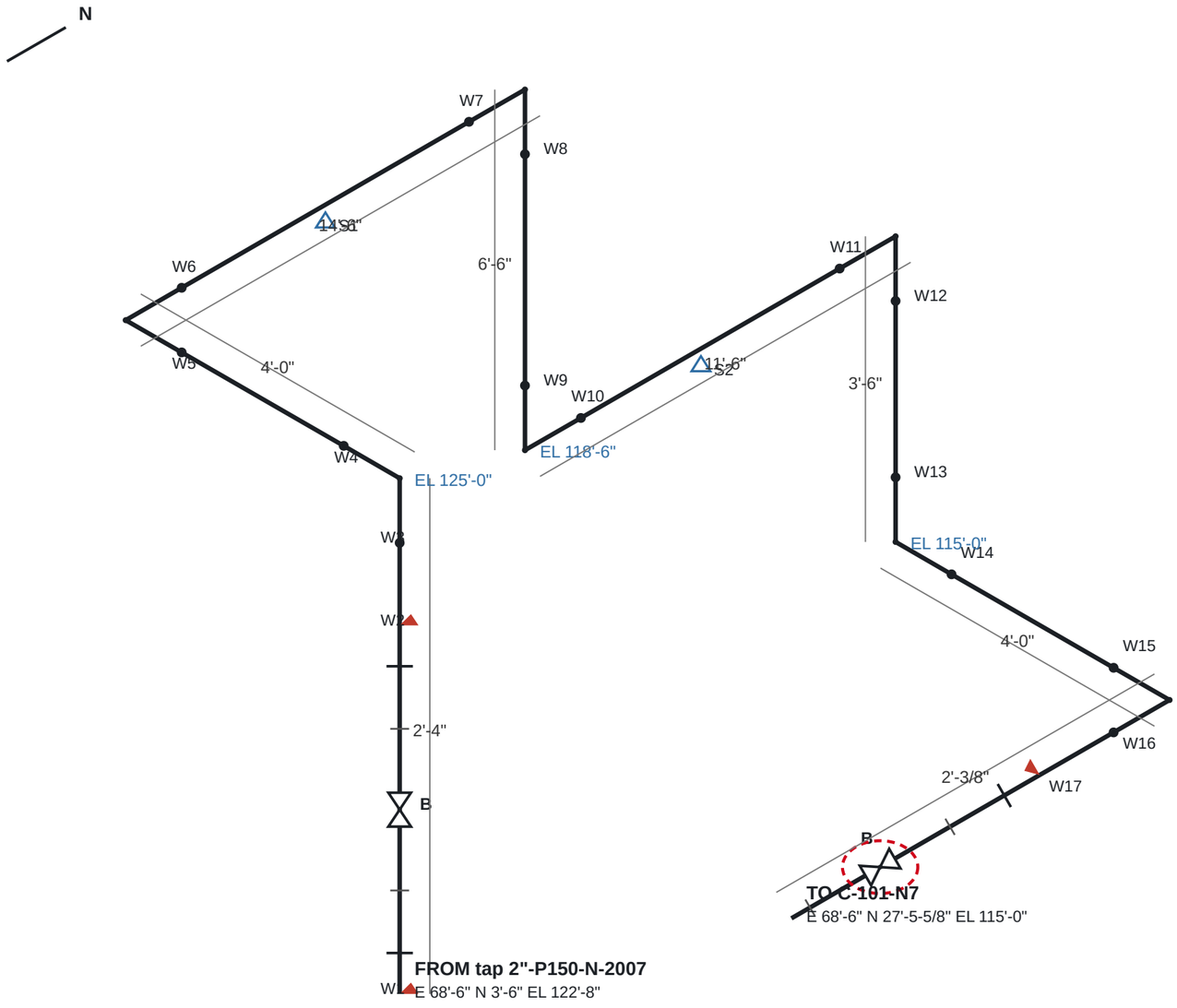
DWG ISO-1"-P150-D-6012 REV 0 WO-200-01

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Isometric ISO-1"-P150-N-3013

ISOMETRIC — 1"-P150-N-3013 — Internal inspection of vertical vessel C-101



BILL OF MATERIALS

1	FLANGE WELD-NECK 1" P150 RAISED FACE	3
2	GASKET 1" P150 COMPRESSED SHEET	4
3	VALVE, BALL, FLOATING 1" P150	2
4	STUD SET 4 × 0.5" × 4" WITH 2 NUTS, FOR 1" P150	4
5	PIPE 1" P150 BEVELLED ENDS	43'-1-3/16"
6	ELBOW 90° R = 5/3 OD 1" P150	7

WELDS 17 (3 FIELD) · SUPPORTS 2 · SPOOLS 4

W1F W2F W3 W4 W5 W6 W7 W8 W9 W10
W11 W12 W13 W14 W15 W16 W17F

1"-P150-N-3013

CLASS G150 — Air and gases
SERVICE N · 1" · P150 · DESIGN 150 psi
tap 2"-P150-N-2007 → C-101-N7
PIPE 43'-1-3/16" · WEIGHT 123 lb

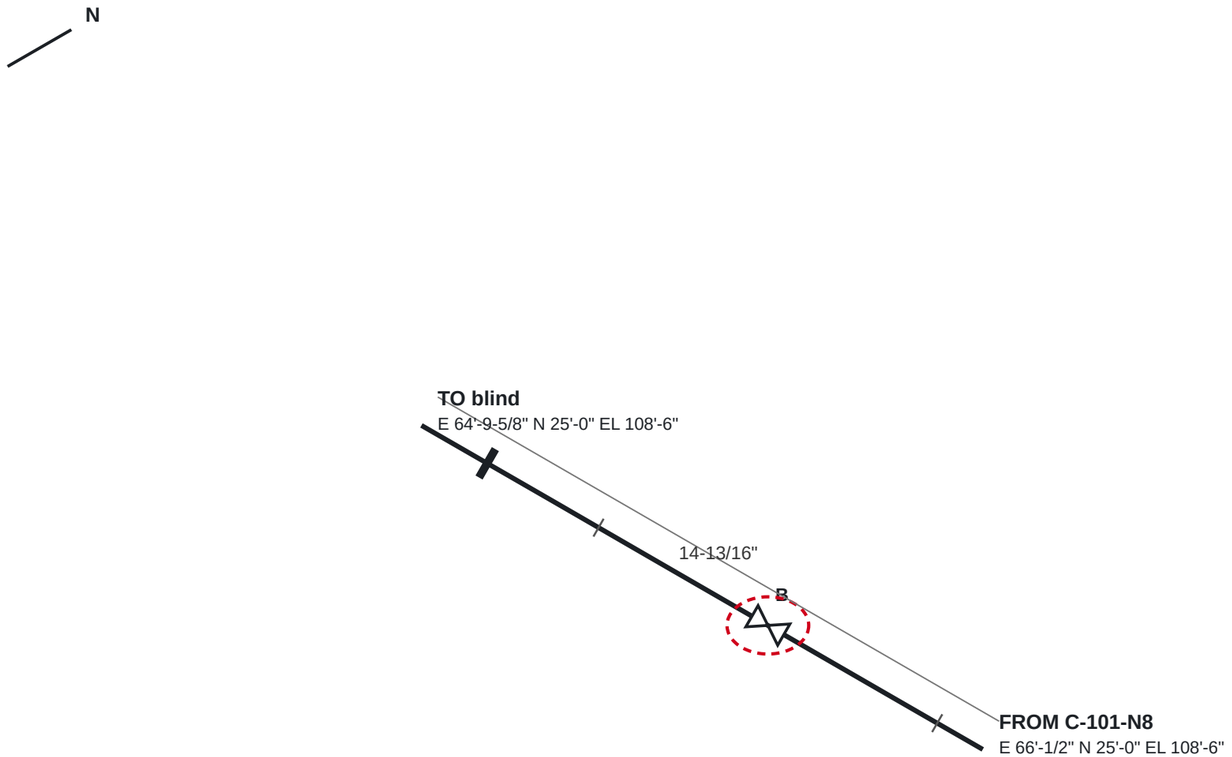
DWG ISO-1"-P150-N-3013 REV 0 WO-200-01

SmartWeld AI · not to scale · E lower right, N upper right, EL up (SW-ISO-1)

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Isometric ISO-3/4"-P300-I-6013

ISOMETRIC — 3/4"-P300-I-6013 — Internal inspection of vertical vessel C-101



BILL OF MATERIALS

1	GASKET 3/4" P300 COMPRESSED SHEET	2
2	VALVE, BALL, FLOATING 3/4" P300	1
3	STUD SET 4 × 0.5" × 4-1/4" WITH 2 NUTS, FOR 3/4" P300	2
4	FLANGE BLIND 3/4" P300 RAISED FACE	1

WELDS 0 (0 FIELD) · SUPPORTS 0 · SPOOLS 0

3/4"-P300-I-6013

CLASS H300 — Hydrocarbon process
SERVICE I · 0.75" · P300 · DESIGN 300 psi
C-101-N8 → blind
PIPE 0.000" · WEIGHT 32 lb

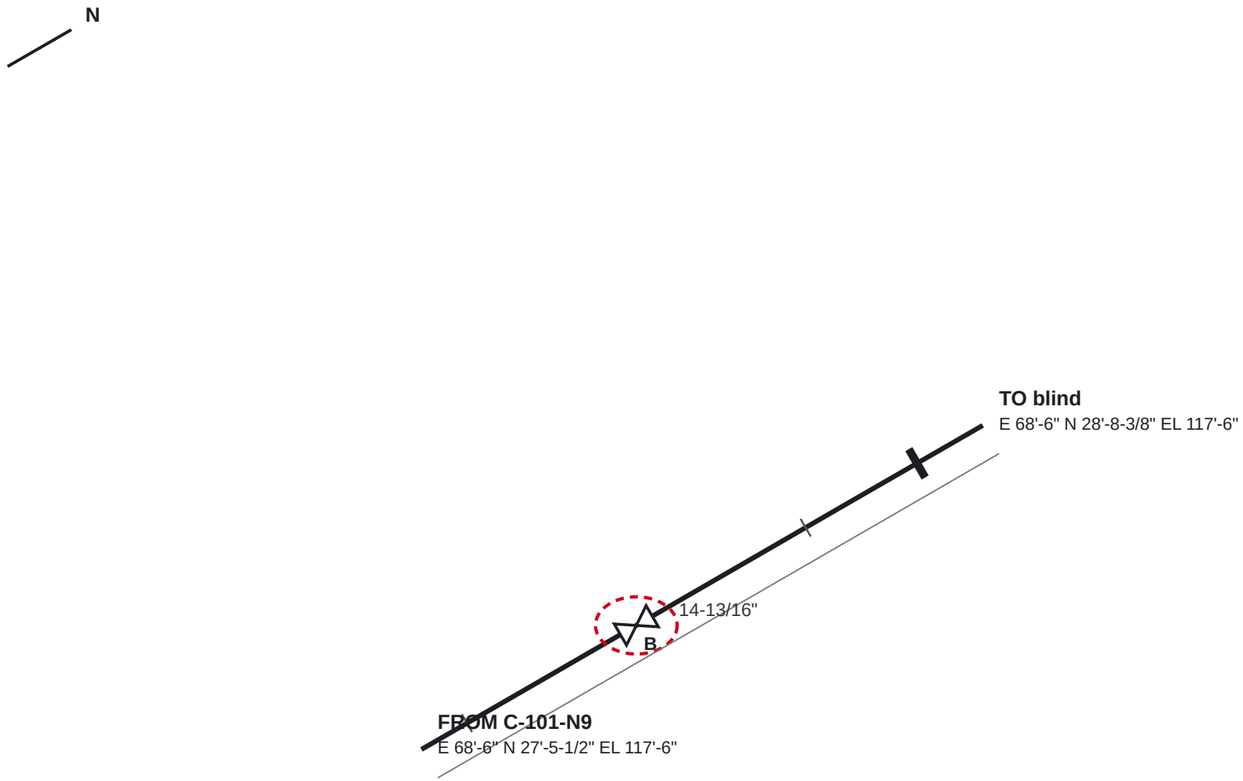
DWG ISO-3/4"-P300-I-6013 REV 0 WO-200-01

SmartWeld AI · not to scale · E lower right, N upper right, EL up (SW-ISO-1)

DEMONSTRATION ONLY — NOT A QUALIFIED PROCEDURE

Isometric ISO-3/4"-P300-I-6014

ISOMETRIC — 3/4"-P300-I-6014 — Internal inspection of vertical vessel C-101



BILL OF MATERIALS

1	GASKET 3/4" P300 COMPRESSED SHEET	2
2	VALVE, BALL, FLOATING 3/4" P300	1
3	STUD SET 4 × 0.5" × 4-1/4" WITH 2 NUTS, FOR 3/4" P300	2
4	FLANGE BLIND 3/4" P300 RAISED FACE	1

WELDS 0 (0 FIELD) · SUPPORTS 0 · SPOOLS 0

3/4"-P300-I-6014

CLASS H300 — Hydrocarbon process
SERVICE I · 0.75" · P300 · DESIGN 300 psi
C-101-N9 → blind
PIPE 0.000" · WEIGHT 32 lb

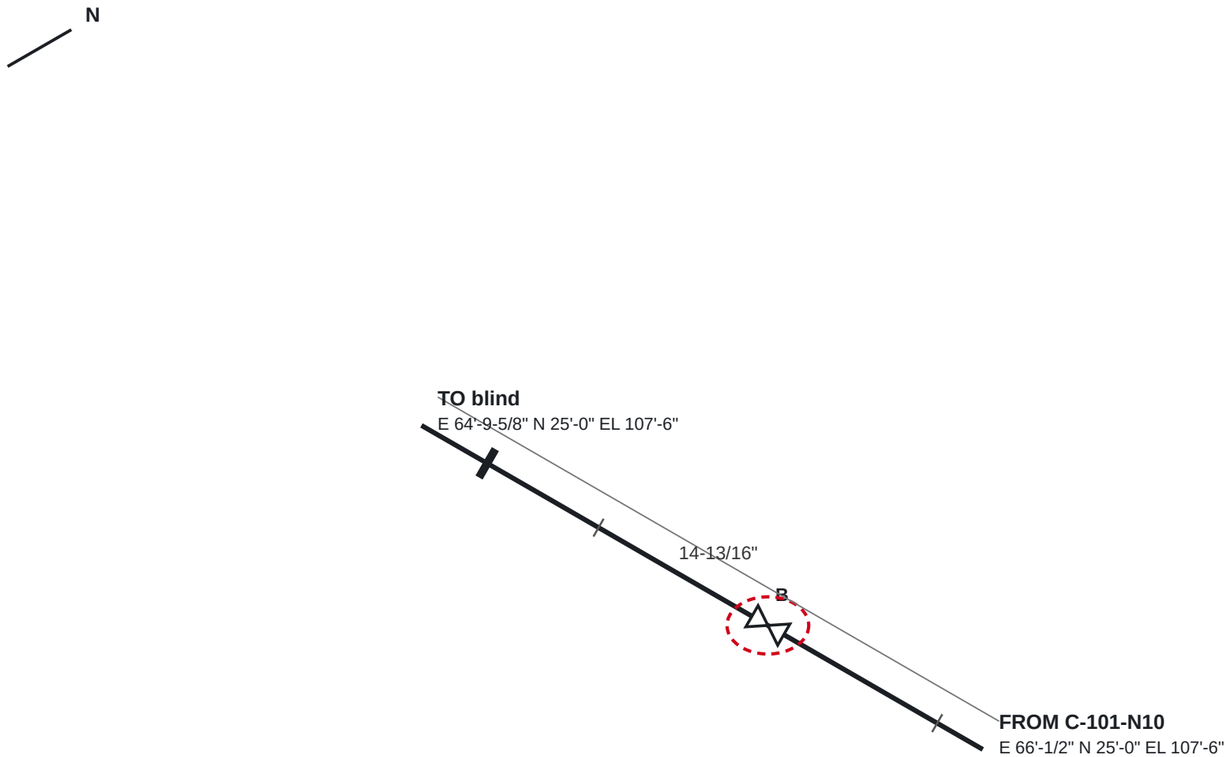
DWG ISO-3/4"-P300-I-6014 REV 0 WO-200-01

SmartWeld AI · not to scale · E lower right, N upper right, EL up (SW-ISO-1)

DEMONSTRATION ONLY — NOT A QUALIFIED PROCEDURE

Isometric ISO-3/4"-P300-SP-6015

ISOMETRIC — 3/4"-P300-SP-6015 — Internal inspection of vertical vessel C-101



BILL OF MATERIALS

1	GASKET 3/4" P300 COMPRESSED SHEET	2
2	VALVE, BALL, FLOATING 3/4" P300	1
3	STUD SET 4 × 0.5" × 4-1/4" WITH 2 NUTS, FOR 3/4" P300	2
4	FLANGE BLIND 3/4" P300 RAISED FACE	1

WELDS 0 (0 FIELD) · SUPPORTS 0 · SPOOLS 0

3/4"-P300-SP-6015

CLASS H300 — Hydrocarbon process
SERVICE SP · 0.75" · P300 · DESIGN 300 psi
C-101-N10 → blind
PIPE 0.000" · WEIGHT 32 lb

DWG ISO-3/4"-P300-SP-6015 REV 0 WO-200-01

SmartWeld AI · not to scale · E lower right, N upper right, EL up (SW-ISO-1)

DEMONSTRATION ONLY — NOT A QUALIFIED PROCEDURE